HW 0303

chapter 2

Question 1.

414707002 簡心怡

a

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i=5$	$\sum y_i=10$	$\sum(x_i-\bar{x})=0$	$\sum(x_i-\bar{x})^2=10$	$\sum(y_i-\bar{y})=0$	$\sum(x-\bar{x})(y-\bar{y})=8$

平方

乘積

$\bar{x}=1$, $\bar{y}=2$

- 差距的大小
- 兩者之間的關係
- 一起變大or變小

for 斜率、相關性

b. OLS estimators

$$b_2=\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} \quad (2.7) \qquad b_1=\bar{y}-b_2\bar{x} \quad (2.8)$$

$$\hat{y}=1.2+0.8x$$

$$b_2=\frac{8}{10}=0.8 \qquad b_1=2-0.8\times 1=1.2$$

c

x	y	x^2	xy
3	4	9	12
2	2	4	4
1	3	1	3
-1	1	1	-1
0	0	0	0

$(x-\text{平均})^2$ 平方總合 - 平均的平方

- $\sum(x-\bar{x})^2=\sum x_i^2-N\bar{x}^2$

$\Rightarrow 10=15-5$

- $\sum(x_i-\bar{x})(y_i-\bar{y})=\sum x_iy_i-N\bar{x}\bar{y}$

$\Rightarrow 8=18-10$

$N=5$ $\sum_{i=1}^{5}x_i^2=15$ $\sum_{i=1}^{5}x_iy_i=18$

$N\bar{x}^2=5\cdot(1)^2=5$

$N\bar{x}\bar{y}=5\cdot1\cdot2=10$

d. $\hat{y} = 1.2 + 0.8x$ $\hat{e} = y - \hat{y}$

残差

x_i	Y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2.	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0

$\sum x_i = 5$ $\sum Y_i = 10$ $\sum \hat{y}_i = 10$ $\sum \hat{e}_i = 0^*$ $\sum \hat{e}_i^2 = 3.6$ $\sum x_i \hat{e}_i = 0^*$

残差平均0 残差与X不相关

1) 样本变异数

$$S_y^2 = \frac{\sum (y_i - \bar{y})^2}{N-1} = \frac{(4-2)^2 + (2-2)^2 + (3-2)^2 + (1-2)^2 + (0-2)^2}{5-1} = \frac{10}{4} = 2.5$$

$$S_x^2 = \frac{\sum (x_i - \bar{x})^2}{(N-1)} = \frac{10}{5-1} = 2.5$$

2) 样本共变异数

$$S_{xy} = \frac{\sum_{i=1}^{N} (y_i - \bar{y})(x_i - \bar{x})}{(N-1)} = \frac{8}{4} = 2$$

3) 样本相关係数

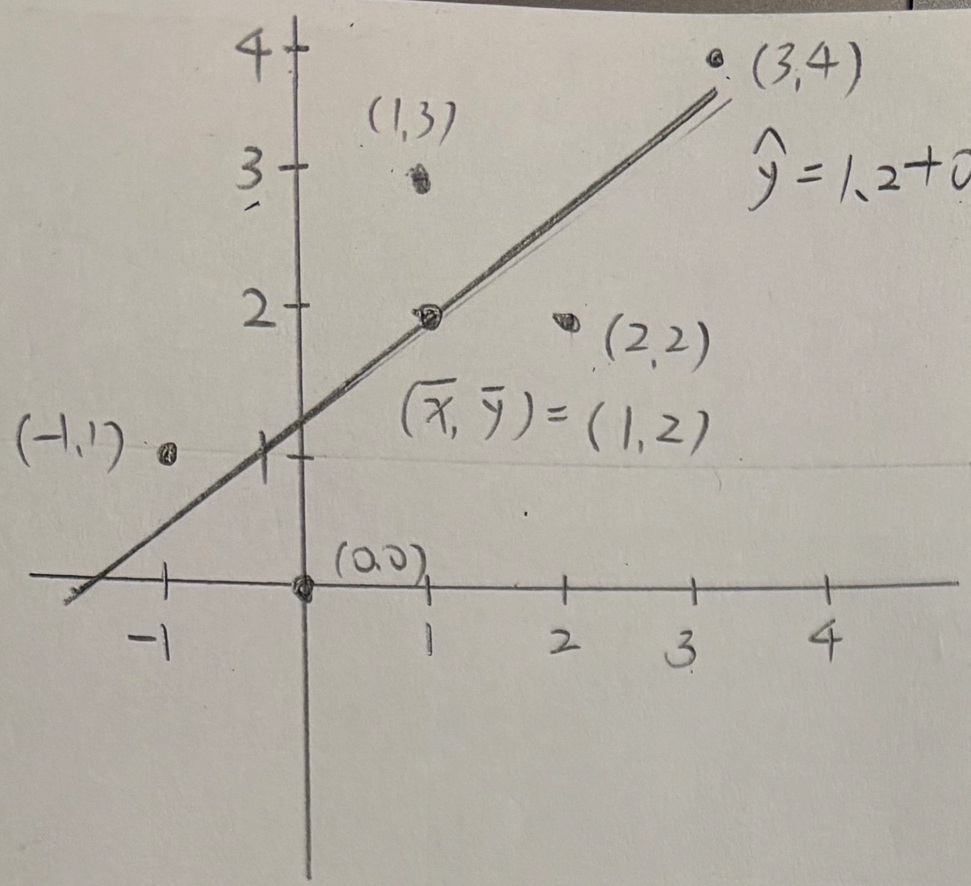
$$r_{xy} = \frac{S_{xy}}{(S_x S_y)} = \frac{2}{\sqrt{2.5} \times \sqrt{2.5}} = \frac{2}{2.5} = 0.8$$

4) 变异係数

$$CV_x = 100 (S_x / \bar{x}) = 100 \times \frac{\sqrt{2.5}}{1} \approx 100 \times 1.5811 \approx 158.11\%$$

X中位数：3, 2, 1, -1, 0 50th percentile, of $x = 1$

e



$\hat{y} = 1.2 + 0.8x$

OLS：求到線的距離平方總合最小

f $(\bar{x}, \bar{y}) = (1,2)$ 迴歸線通過 $(\bar{x}, \bar{y}) = (1,2)$

g $\bar{y} = b_1 + b_2\bar{x}$

$2 = 1.2 + 0.8 \times 1$

h $\overline{\hat{y}} = \bar{y}$ $\Sigma \hat{y}_i = 10$ $\overline{\hat{y}} = \frac{10}{5} = 2$ $\overline{\hat{y}} = \bar{y} = 2$

i $\hat{\sigma}^2$ = OLS 誤差變異估計

$$\hat{\sigma}^2 = \frac{\Sigma e_i^2}{N-2}$$ ← 估計2個參數 (b_1, b_2) 自由度 $N-K$

$$= \frac{3.6}{5-2}$$

$$= 1.2$$

j $\widehat{var}(b_2|X) =$ $Var(b_2|X) = \frac{\sigma^2}{\Sigma(x_i - \bar{x})^2}$

$$\widehat{var}(b_2) = \frac{\hat{\sigma}^2}{\Sigma(x_i - \bar{x})^2}$$

$$= \frac{1.2}{10} = 0.12$$

$Se(b_2) = \sqrt{0.12}$

標準誤 $= 0.346$

CH2_2.22

<table>
<tr>
<td>a</td>
<td>

```
> #2.22a
> #載入資料套件
> library(POE5Rdata)
> #建立迴歸模型
> #linear model (線性迴歸)
> #data = star5_small 資料來源
> mod1<-lm(totalscore~small,data=star5_small)
> smod1<-summary(mod1)
> smod1

Call:
lm(formula = totalscore ~ small, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max
-191.63  -50.63   -7.12   42.37  324.38

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  918.628      2.536 362.260   <2e-16 ***
small          9.989      4.611   2.166   0.0305 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.36 on 1198 degrees of freedom
Multiple R-squared:  0.003903,  Adjusted R-squared:  0.003071
F-statistic: 4.694 on 1 and 1198 DF,  p-value: 0.03047
```

</td>
<td>小班的估計係數(9.989)表明，小班的平均得分比普通班學生高約 10 分，而此差異在5%的顯著水準上具有統計意義(P=0.0305)。
$\mathrm{TOTALSCORE_i} = 918.628 + 9.989SMALL_i + E_i$</td>
</tr>
<tr>
<td>b(1)</td>
<td>

```
> mod2<-lm(readscore~small,data= star5_small)
> smod2<-summary(mod2)
> smod2

Call:
lm(formula = readscore ~ small, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max
-76.123 -21.123  -3.356  15.877 192.877

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  434.123      1.044 416.021  < 2e-16 ***
small          5.466      1.897   2.881  0.00403 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.19 on 1198 degrees of freedom
Multiple R-squared:  0.006882,  Adjusted R-squared:  0.006053
F-statistic: 8.301 on 1 and 1198 DF,  p-value: 0.004032
```

</td>
<td>小班學生的估計係數(5.466)表明，小班學生的閱讀平均得分比普通班級學生高約 5.5 分，而此差異在 1%的顯著水準上具有統計意義(P=0.00403) 。
$\mathrm{READSCORE_i} = 434.123 + 5.466SMALL_i + E_i$</td>
</tr>
<tr>
<td>b(2)</td>
<td>

```
> mod3<-lm(mathscore~small,data=star5_small)
> smod3<-summary(mod3)
> smod3

Call:
lm(formula = mathscore ~ small, data = star5_small)

Residuals:
     Min       1Q   Median       3Q      Max
-145.505  -35.028   -6.505   28.495  141.495

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  484.505      1.677 288.884   <2e-16 ***
small          4.522      3.049   1.483    0.138
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.52 on 1198 degrees of freedom
Multiple R-squared:  0.001832,  Adjusted R-squared:  0.0009992
F-statistic: 2.199 on 1 and 1198 DF,  p-value: 0.1383
```

</td>
<td>小班學生的估計係數(4.522)表明，小班學生的數學平均得分比普通班級學生高約 4.5 分，而此差異不具有統計意義(P=0.138) 。
$\mathrm{MATHSCORE_i} = 484.505 + 4.522SMALL_i + E_i$</td>
</tr>
</table>

b 估計結果表明，小班制的學生在閱讀和數學方面的表現都更好。然而這種影響在閱讀成績方面具有統計意義，但在數學成績方面則不顯著。

<table>
<tr>
<td>c</td>
<td>

```
> mod4<-lm(totalscore~aide,data=star5_small)
> smod4<-summary(mod4)
> smod4

Call:
lm(formula = totalscore ~ aide, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max
-195.14  -52.14   -6.95   40.95  330.86

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  922.145      2.640 349.250   <2e-16 ***
aide          -1.396      4.437  -0.315    0.753
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 73.5 on 1198 degrees of freedom
Multiple R-squared:  8.267e-05, Adjusted R-squared:  -0.000752
F-statistic: 0.09904 on 1 and 1198 DF,  p-value: 0.753
```

</td>
<td>助教的估計係數(-1.396)表明，在有助教的普通班級中，學生的平均得分比沒助教的班級低約 1.4 分，但這種差異在統計上並不顯著(P=0.753)。
$\mathrm{TOTALSCORE_i} = 922.145 - 1.396 AIDE_i + E_i$</td>
</tr>
<tr>
<td>d(1)</td>
<td>

```
> mod5<-lm(readscore~aide,data=star5_small)
> smod5<-summary(mod5)
> smod5

Call:
lm(formula = readscore ~ aide, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max
-77.908 -21.536  -2.908  15.464 191.092

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 435.9084     1.0882 400.585   <2e-16 ***
aide         -0.3719     1.8285  -0.203    0.839
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 30.29 on 1198 degrees of freedom
Multiple R-squared:  3.453e-05, Adjusted R-squared:  -0.0008002
F-statistic: 0.04137 on 1 and 1198 DF,  p-value: 0.8389
```

</td>
<td>助教的估計係數(-0.3719)表明，在有助教的普通班級中，閱讀平均得分比普通班級學生低約 0.4 分，但這種差異在統計上並不顯著(P=0.839)。
$\mathrm{READSCORE_i} = 435.9084 - 0.3719 AIDE_i + E_i$</td>
</tr>
<tr>
<td>d(2)</td>
<td>

```
> mod6<-lm(mathscore~aide,data=star5_small)
> smod6<-summary(mod6)
> smod6

Call:
lm(formula = mathscore ~ aide, data = star5_small)

Residuals:
     Min       1Q   Median       3Q      Max
-147.236  -32.236   -7.212   27.788  140.788

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  486.236      1.744 278.731   <2e-16 ***
aide          -1.024      2.931  -0.349    0.727
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 48.56 on 1198 degrees of freedom
Multiple R-squared:  0.0001019, Adjusted R-squared:  -0.0007327
F-statistic: 0.1221 on 1 and 1198 DF,  p-value: 0.7268
```

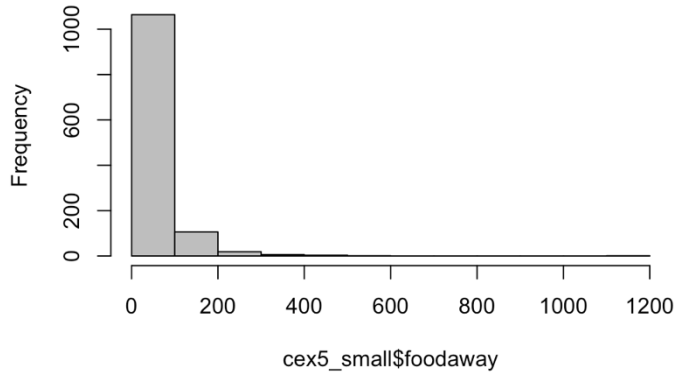
</td>
<td>助教的估計係數(-1.024)表明，在有助教的普通班級數學平均得分比沒助教的普通班級低約 1.024 分，但此差異不具有統計意義(P=0.727)。
$\mathrm{MATHSCORE_i} = 486.236 - 1.024 AIDE_i + E_i$</td>
</tr>
</table>

估計結果表明，在沒有助教的普通班級的學生閱讀和數學成績都優於有助教的普通班級，但這兩項結果在統計上都不顯著。

CH2_2.25

a

```
#直方圖
hist(cex5_small$foodaway,col = "grey")
```




```
> #敘述統計
> summary(cex5_small$foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   12.04   32.55   49.27   67.50 1179.00
```

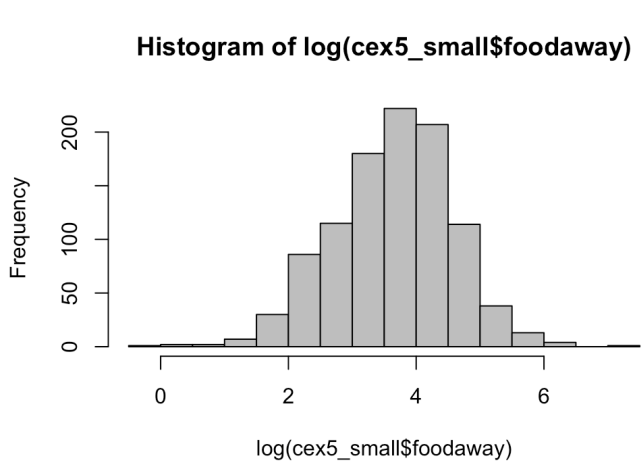
- ✓ mean =49.27
- ✓ median=32.55
- ✓ 25^{th} =12.04
- ✓ 75^{th}= 67.50

b

```
> #條件篩選
> #有大學教育
> summary(cex5_small$foodaway[cex5_small$advanced==1])
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   21.67   48.15   73.15   90.00 1179.00
> #有進階教育
> summary(cex5_small$foodaway[cex5_small$college==1])
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   14.44   36.11   48.60   68.67  416.11
> #沒大學 + 沒進階教育
> summary(cex5_small$foodaway[cex5_small$advanced==0&cex5_small
$college==0])
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00    9.63   26.02   39.01   52.65  437.78
```

	平均數	中位數
高階學歷	73.13	48.15
大學學歷	48.60	36.11
沒有高階&大學學歷	39.01	26.02

c




```
> #c
> hist(log(cex5_small$foodaway),col = 'grey')
> summary(log(cex5_small$foodaway))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   -Inf   2.488   3.483    -Inf   4.212   7.072
```

由於 FOODAWAY 中至少有一個資料為 0，而 0 的對數不存在，因此 ln (FOODWAY) 的觀測值少於 FOODWAY。

<table>
<tr>
<td>d</td>
<td>

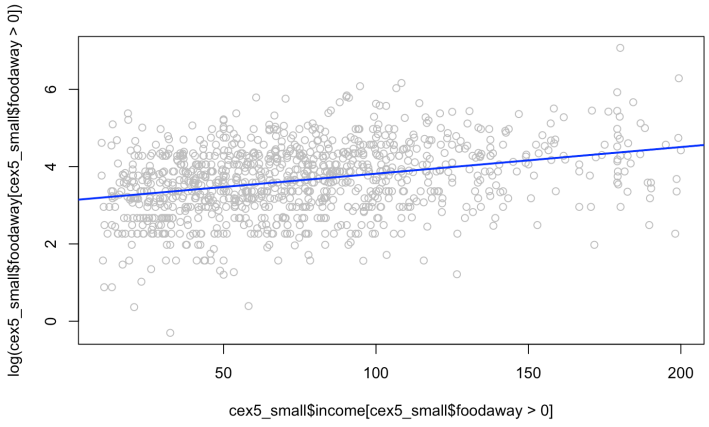
```
> #d
> #log(0) 不存在
> mod1<-lm(log(foodaway)~income,data=cex5_small[cex5_small$foodaway>0,])
> smod1<-summary(mod1)
> smod1

Call:
lm(formula = log(foodaway) ~ income, data = cex5_small[cex5_small$foodaway >
    0, ])

Residuals:
    Min      1Q  Median      3Q     Max
-3.6547 -0.5777  0.0530  0.5937  2.7000

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 3.1293004  0.0565503   55.34   <2e-16 ***
income      0.0069017  0.0006546   10.54   <2e-16 ***
---
Signif. codes:
0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8761 on 1020 degrees of freedom
Multiple R-squared:  0.09826,   Adjusted R-squared:  0.09738
F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

</td>
<td>

$\ln(FOODAWAY) = 3.1293004 + 0.0069017INCOME + e$

在 log(Y)模型中，估計斜率係數 0.0069017 表示，如果每月收入增加 100 美元，每人每月外食支出大約會增加 0.69%

</td>
</tr>
<tr>
<td>e</td>
<td>




```
> #e
> plot(cex5_small$income[cex5_small$foodaway>0],log(cex5_small$foodaway[cex5_small$fooda
way>0]),col="grey")
> abline(mod1,col="blue",lwd=2)
```

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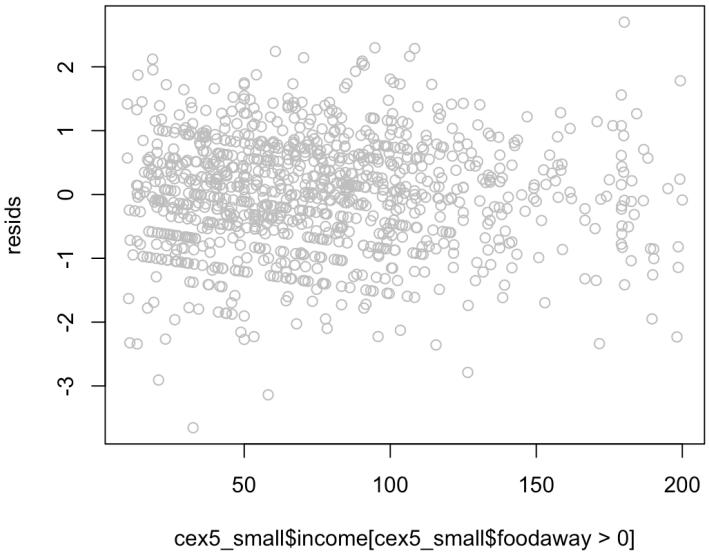
<tr>
<td>f</td>
<td>




```
> #f
> resids<-residuals(mod1)
> plot(cex5_small$income[cex5_small$foodaway>0],resids,col
="grey")
```

</td>
<td>殘差似乎隨機散佈在零周圍，沒有明顯的模式。
較高收入水準的分散率略有增加，這表明可能存在輕微的異質性。</td>
</tr>
</table>